Tessa Frincke

Research Interests

Minor Bodies, Solar System Dynamics, Interstellar Objects, Exoplanets

Education

2025- **Ph.D.**, *Michigan State University, Department of Physics and Astronomy*, East Lansing, MI.
- Ph.D. Advisor: Professor Darryl Seligman

2021-2025 **B.S.**, *University of Toledo*, Toledo, OH., Physics, Concentration Astrophysics, Magna Cum Laude.
- Senior Honors Thesis: "A PAH Investigation of Star Clusters in the Antennae Galaxies NGC 4038/39"
- Thesis Advisor: Professor Rupali Chandar

Teaching Experience

2025- **Graduate Teaching Assistant**, *Michigan State University*, East Lansing, MI.
- Visions of the Universe Laboratory: Leads, plans, and manages lab section, gives short lecture at the beginning of lab, assists non-major students in lab, and grades weekly lab work.
- Studio Physics for Life Scientists I: Holds weekly office hours, assists pre-med students in class, and grades lab reports.

Honors And Awards

2025 Jesup Scott Honors College Medallion, University of Toledo, Toledo, OH

2024 NSF REU, University of Michigan, Ann Arbor, MI

2021-2025 Dean's List for 8 semesters, University of Toledo, Toledo, OH

Leadership and Service

2024-2025 Vice President, Society of Physics Students, University of Toledo, Toledo, OH

Selected Grants & Observing Proposals

2025 Michigan-Dartmouth-MIT Observatory PI Salazar Manzano, A11pl3Z, Chasing the third interstellar object 3I ATLAS, co-I, 10 nights

Observational Experience

09/2025 Remote Observations of 3I/ATLAS, Southern Astrophysical Research (SOAR) Telescope, Cerro Pachón, Chile, 3 half-nights, Lead Observer

✉ frincket@msu.edu

08/2024 Patroclus-Menoetius Occultation Campaign, Southwest Research Institute, Boulder, CO, 4 nights

10/2024 Student Observing Campaign, Lowell Discovery Telescope, Flagstaff, AZ, 1 night

First Author Publications *(ADS Link)*

2. **Frincke, T.** and Seligman, D., et al. "Assessment of the Mass Loss and Radius Change of 3I/ATLAS Based on Observed Production Rates." *MNRAS*, Accepted, 2026. (Link)

1. **Frincke, T.**, Yaginuma, A., et al. "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS." *MNRAS*, Accepted, 2026. (Link)

Other Publications

4. Salazar Manzano, L., Lin, H., Tyler, A., Seligman, D., Adams, F., Gerdes, D., Ruch, T., **Frincke, T.**, and Napier, J. , "Onset of CN Emission in 3I/ATLAS: Evidence for Strong Carbon-Chain Depletion." *ApJL*, Accepted, 2025. (Link)

3. Yaginuma, A., **Frincke, T.**, Seligman, D., Mandt, K, DellaGiustina, D., Peña-Asensio, E., Taylor, A., Nolan, M., "The Feasibility of a Spacecraft Flyby with the Third Interstellar Object 3I/ATLAS from Earth or Mars." *ApJ*, Accepted, 2025. (Link)

2. Seligman, D., Micheli, M., Farnocchia, D. and 44 coauthors including **Frincke, T.**, "Discovery and Preliminary Characterization of a Third Interstellar Object: 3I/ATLAS." *ApJL*, Accepted, 2025. (Link)

1. Salazar Manzano, L., Gerdes, G., and 6 coauthors including **Frincke, T.**, "Prospects of a New L5 Trojan Flyby Target for the Lucy Mission." *PSJ*, Accepted, 2025. (Link)

Invited Presentations

09/2026 **Frincke, T.**, ""3I/ATLAS: Our Most Recent Interstellar Interloper," Oral presentation, Astrophysical and Planetary Science (APS) Friday Lunch Seminar, University of Colorado Boulder, Boulder, CO.

01/2026 **Frincke, T.**, "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS," Oral presentation, Invited Early Career Speaker at the 34th Meeting of the NASA Small Bodies Assessment Group (SBAG), Johns Hopkins University, Baltimore, MD.

Contributed Presentations

09/2026 **Frincke, T.**, "Assessment of the Mass Loss and Radius Change of 3I/ATLAS Based on Observed Production Rates," Oral presentation at the Integrated Science of Comets: From Laboratory Studies of Cometary Materials to Remote Observations Workshop, Lunar and Planetary Institute/Universities Space Research Association, Virtual.

09/2026 **Frincke, T.**, "The Surface Erosion of 3I/ATLAS: Our Most Recent Interstellar Interloper," Oral presentation at Southwest Research Institute Seminar, Southwest Research Institute Boulder, CO.

07/2026 **Frincke, T.**, "Assessment of the Mass Loss and Radius Change of 3I/ATLAS Based on Observed Production Rates," Oral presentation at the Emerging Researchers in Exoplanet Science (ERES) Symposium XI, University of Michigan, Ann Arbor, MI.

07/2026 **Frincke, T.**, "Observability of Planet Forming Materials From Interstellar Comet 3I/ATLAS," Oral presentation at the 15th Asteroid, Comets, Meteors (ACM) Conference, Poznań University of Technology, Poznań, Poland.

11/2025 **Frincke, T.**, "Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS," Poster presentation at the Great Lakes Exoplanet Area Meeting (GLEAM), University of Wisconsin-Madison, Madison, WI.

02/2025 **Frincke, T.**, "Characterization of Star Clusters in the Antennae Galaxies NGC 4038/39," Poster presentation at the Jesup Scott Honors College Open House Dinner, University of Toledo, Toledo, OH.

01/2025 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Poster presentation at Conference for Undergraduate Women and Gender Minorities in Physics (CU*iP), Michigan Technological University, Houghton, MI.

12/2024 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Poster presentation at Undergraduate Research and Creative Activity Exhibition, University of Toledo, Toledo, OH.

11/2024 **Frincke, T.**, Stewart, C., and Beach, K., "Investigation of Star Clusters in the Arp220 Galaxy," Oral presentation at Ritter Symposium, University of Toledo, Toledo, OH.

10/2024 Gerdes, D., Levison, H., Napier, K., Salazar Manzano, L., Lin, J., and **Frincke, T.**, "Spacecraft encounters with resonant populations: an optimal target search strategy," Oral presentation at AAS Division for Planetary Sciences meeting 56, id. 209.04. Bulletin of the American Astronomical Society, Vol. 56, No. 8 e-id 2024n8i209p04

08/2024 **Frincke, T.**, "Simulating and Identifying Encounter-able Hilda Asteroids," Oral presentation at NSF REU Symposium, University of Michigan, Ann Arbor, MI.